

Nhat-Phat Nguyen

Final-Year AI / Computer Vision Researcher

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Research Interests

3D Reconstruction & Neural Rendering (3D Gaussian Splatting) · Medical Image Analysis
· Deep Learning · Object Detection & Edge AI

Education

2023–Present **B.Eng. in Information Technology**, *HCMC University of Technology and Engineering (HCM-UTE)*, Ho Chi Minh City, Vietnam, GPA: 3.65 / 4.0
Concentration in Artificial Intelligence and Computer Vision. Recipient of the HCM-UTE Talented Student Scholarship (2024) and Academic Excellence Scholarship (2024).

Publications

Conferences

Nhat-Phat Nguyen, Quoc-Hung Pham, Van-Dung Hoang[†].
“Adaptive Density-Based Geometry Compression for Efficient 3D Gaussian Splatting.”
The 8th International Conference on Green Technology and Sustainable Development (**GTSD 2026**), Springer Track. **(Accepted)**
Introduced a geometry-aware compression method using adaptive density estimation to remove spatially redundant Gaussian primitives while preserving critical structures, achieving 30% smaller model size with no quality loss across 11 scenes on two standard benchmarks.

Preprints

Nhat-Phat Nguyen, Quoc-Hung Pham, Van-Dung Hoang[†].
“Self-Supervised CT Through-Plane Interpolation via Residual 3D Gaussian Splatting.”
Independent Research, 2026. **(Preprint)**
A self-supervised approach reconstructing high-resolution volumetric CT stacks from sparse axial slices without paired supervision, using residual 3D Gaussian Splatting.

[†] Corresponding author / research supervisor. **Bold** indicates self.

Research & Selected Projects

2026 **Real-Time Construction Site Safety Monitoring System**, *HCM-UTE – Faculty of Advanced Education*, AI/ML Lead
Designed a zone-aware PPE compliance detection system on edge hardware (Jetson Xavier NX). YOLO26n-based model achieving 0.677 mAP@0.5 at 131 FPS with 99.93% bandwidth reduction versus cloud streaming. Awarded 3rd Prize, Edge Computing Innovation 2026.

- 2024–2025 **Advanced Retinal Vessel Analysis Using Deep Learning**, *HCM-UTE University Research Program, Lead Researcher*
Conducted a deep-learning research project segmenting retinal blood vessels from medical images. Applied CNNs and image processing for high-resolution pixel-level medical image analysis. Rated “Good”; presented at Euréka XXVII 2025 (VNU-HCM).
- 2024 **Cassava Leaf Disease Classification**, *Independent Project*
Developed a MobileNetV3 classifier with k-fold cross-validation and data augmentation, achieving over 90% validation accuracy, with a desktop GUI interface.

Awards & Honors

- Jul 2026 **1st Prize** – Project Showcase 2026, Academic Pathways Track (HCM-UTE)
- Jul 2026 **3rd Prize** – Edge Computing Innovation 2026 (Faculty of Advanced Education, HCM-UTE)
- Apr 2026 Outstanding Student in Scientific Research – FAEPRIME, HCM-UTE (AY 2024–2025)
- Jan 2026 Student of Five Merits – **Ho Chi Minh City Level**
- Mar 2026 Student of Five Merits – University Level (HCM-UTE)
- Oct 2025 Euréka XXVII 2025 – Student Scientific Research Prize (Vietnam National University HCM)
- Dec 2024 HCM-UTE Talented Student Scholarship
- Dec 2024 Academic Excellence Scholarship – Good Standing (Conduct score: 100/100)
- Jan 2024 Young Leader Face of Ho Chi Minh City 2023 (HCMC Youth Union)

International Academic Exchange

- Jan 2026 **BMW Program – Digital Twin & AI-Based Industrial Simulation**, *University of Ulsan, South Korea*
Inbound academic exchange on digital twin technology and AI-based industrial simulation; collaborated with international students.

Technical Skills

Languages	Python, C++, C#, SQL	ML/DL Frameworks	PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV
Tools & MLOps	GitHub, Docker, CUDA, Kaggle, Colab	Languages (spoken)	Vietnamese (Fluent), English (IELTS 6.0)

Leadership & Service

- 2023–Present Youth Union Secretary, Class 23110FIE1, HCM-UTE – coordinated student activities and community engagement
- 2024–2025 Volunteer: Spring Volunteer Campaigns, Exam Season Support; volunteer social work score 99/100